UNIVERSITAT DE BARCELONA

ICCUB

Institut de Ciències del Cosmos
UNIVERSITAT DE BARCELONA

Master in Astrophysics, Particle Physics, and Cosmology

Academic year 2025-2026

Star Formation and Structure

Fall semester: Mon, Tue, Wed, Thur, 16:10h – 17:10h
Room V12M

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Nadia Blagorodnova (nblago@fqa.ub.edu)











Lines of Research in the Star Formation Group

- High-angular resolution observations of the first stages of stellar evolution
- Outflows, jets, and accretion disks in low- and high-mass young stellar objects
- Astrochemistry

In collaboration:

CSIC Institute of Space Sciences

EXCELENCIA MARÍA DE MAEZTU









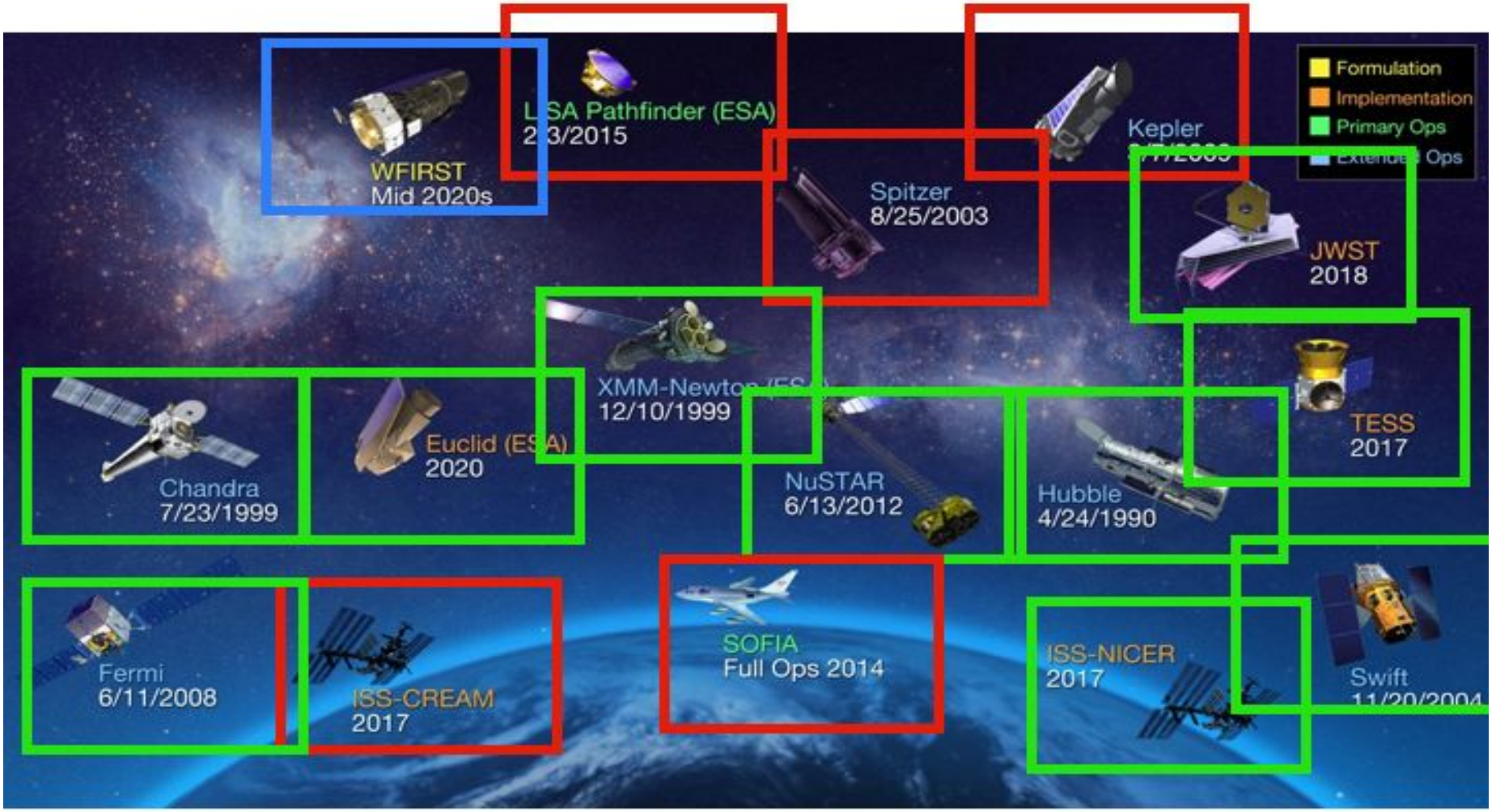
Lines of Research in the Stellar Evolution Group

- Observational study of different stages of binary stellar evolution:
  - Interacting binary systems and variability studies in large time-domain surveys.
  - Astrophysical transients resulting from binary interaction: stellar mergers and interacting supernovae.
  - Binary interaction remnants: geometries and dust formation
  - Connection of binary remnants to GW sources

MULTI-WAVELENGTH OBSERVATIONS





How do stars form and evolve?

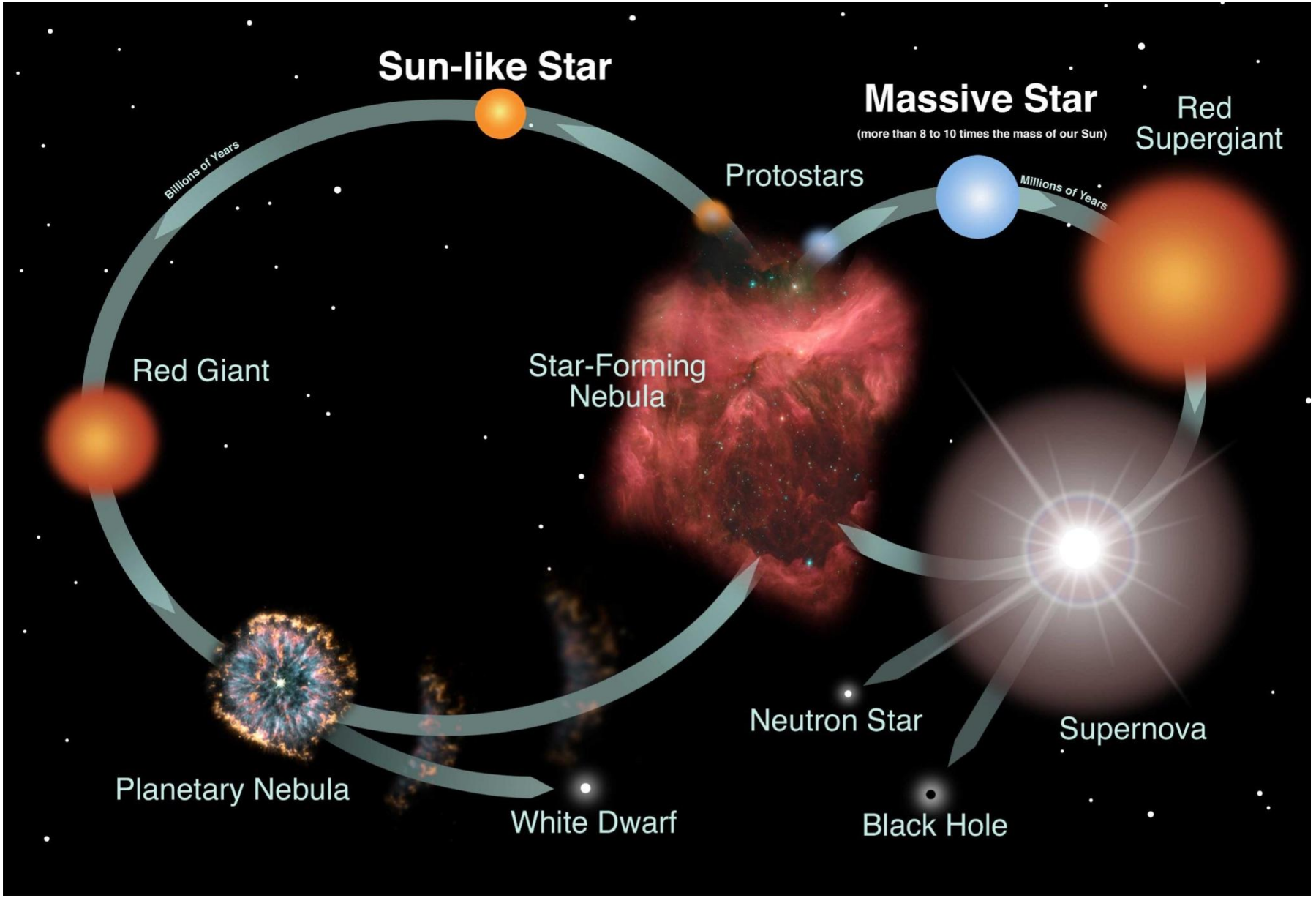
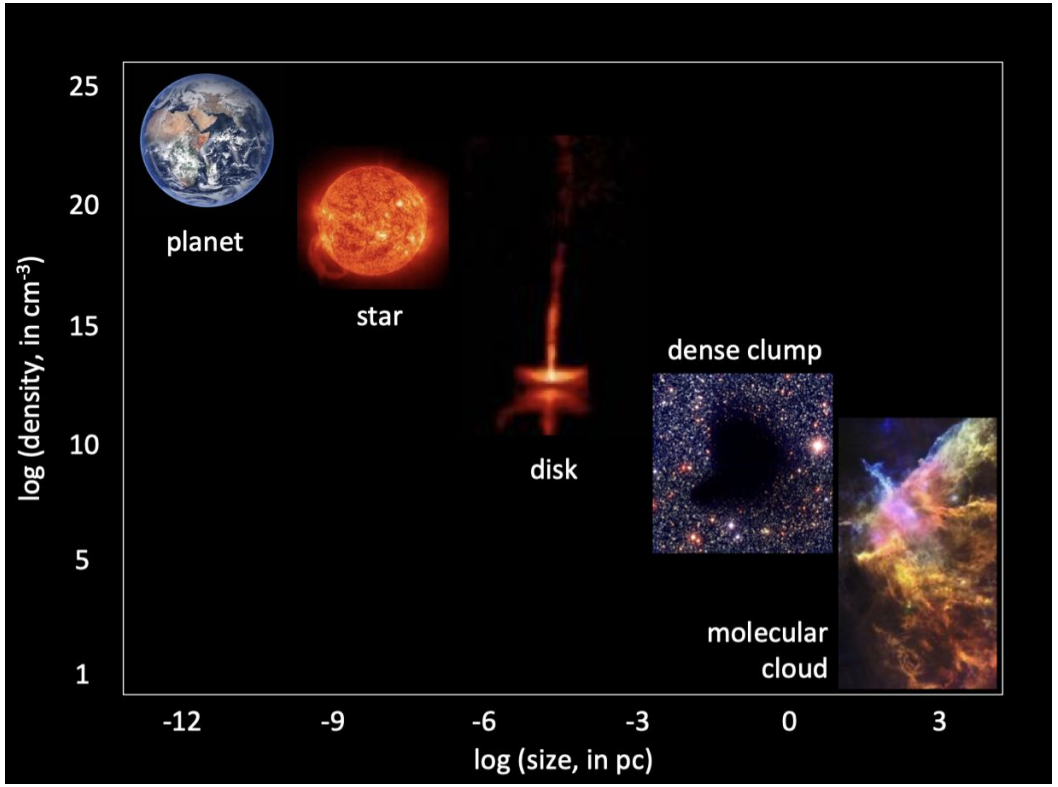


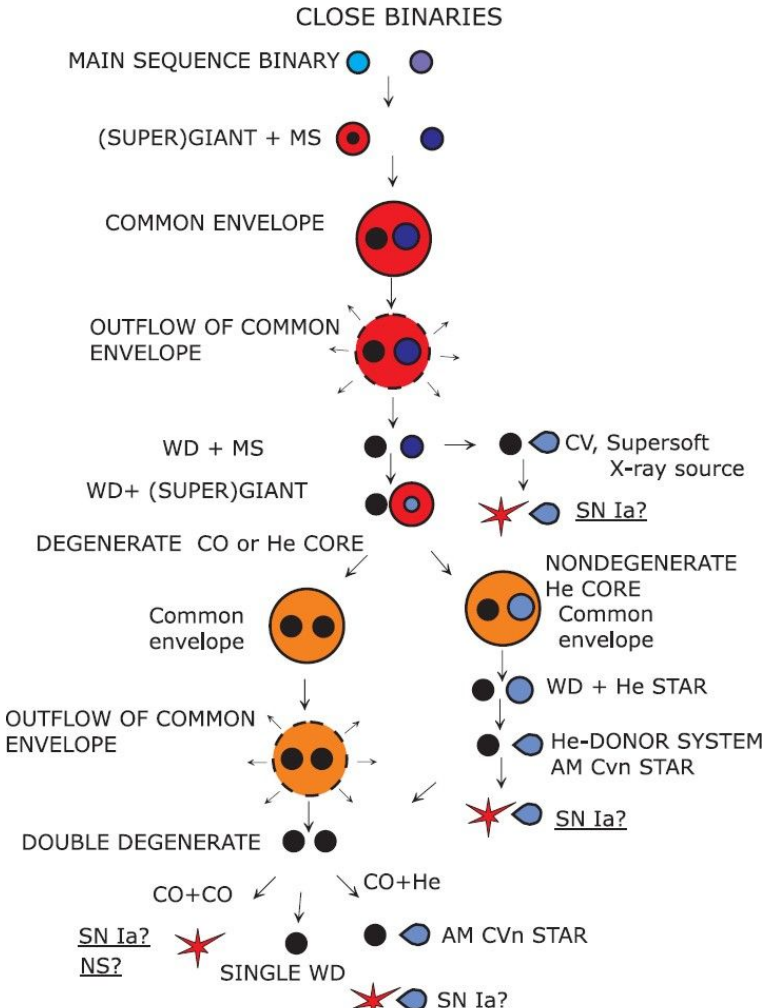
Image Credit: *NASA and the Night Sky Network.*

OBJECTIVES

To understand the physics of the ISM, the star formation process in our Galaxy and the pre-main sequence evolution of stars with different masses



To understand the evolutionary channels of binary stars, and their final fate.



TEACHING BLOCKS

1. Introduction to the ISM — Gemma

- The history of the ISM and its structure
- Interstellar dust
- Interstellar gas

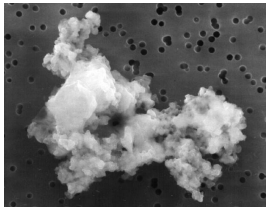
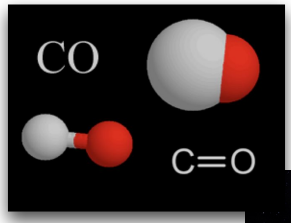


Image Credit: NASA

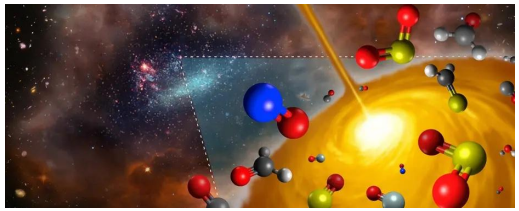




Credit: R. Friesen, Dunlap Institute/J. Pineda, MPIP /GBO /AUI /NSF /NASA

2. Star-forming regions — Gemma

- Molecular clouds and energy balance
- Cloud collapse, fragmentation and the IMF
- Low- and high-mass star formation
- Astrochemistry



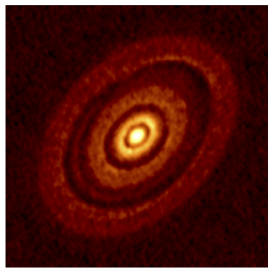
Credit: ALMA Observatory

3. Young Stellar Objects — Rosario

- Spectral Energy Distribution
- Pre-main sequence evolution
- T Tauri and Ae/Be stars
- Interaction of YSOs with their environment: jets, HH objects and molecular outflows
- Accretion and supersonic ejection processes
- Protoplanetary disks — Gemma



Image Credit: ESA/Webb, NASA, CSA, Tom Ray (Dublin)



Credit: ALMA (ESO/NAOJ/ NRAO)

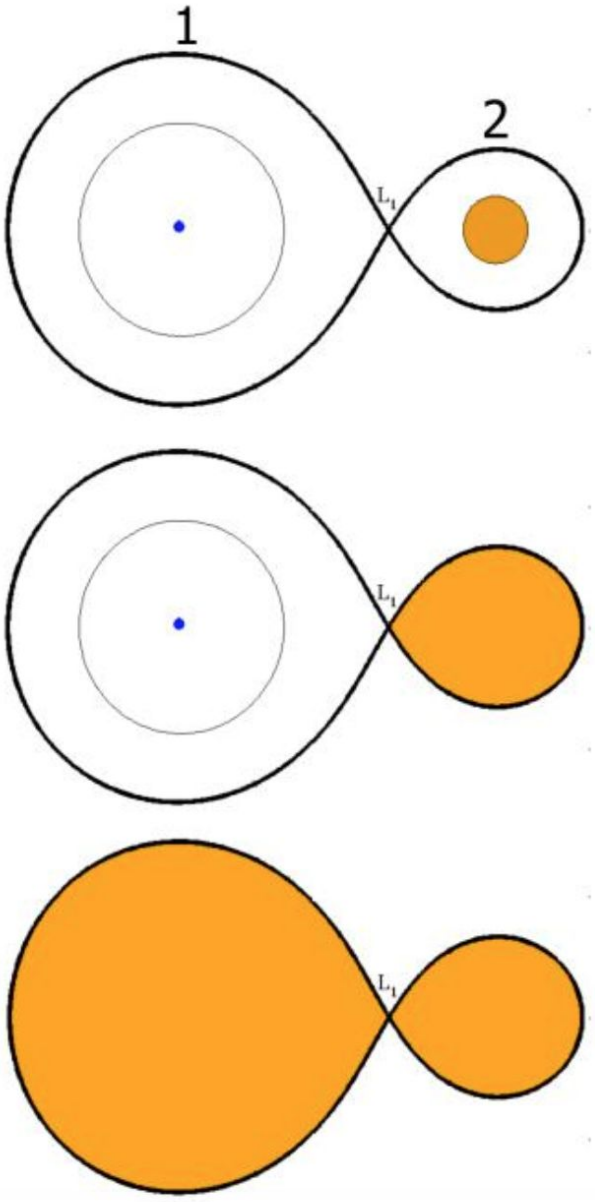
TEACHING BLOCKS

4. Stellar Structure and Evolution Nadia

- Stellar structure
- Energy generation and transport
- Stellar timescales
- Single stellar evolution: low- and high-mass

5. Binary systems Nadia

- Roche lobe geometry
- Types of binaries
- Mass transfer and common envelope evolution
- Binary remnants and GW sources
- Astrophysical transients from binaries



TEACHING BLOCKS

6. Observational techniques and modelling tools

- Radio astronomy Gemma
- Optical and near-infrared astronomy Rosario
- MESA stellar evolution code Nadia





Practical applications

Gemma

Proposal writing



Basic concepts on AOT, calibration and imaging using CASA
Guided projects using real interferometric ALMA data

ALMA Observing Tool

CASA
Common Astronomy Software Applications

Nadia

Evolution of a binary star in MESA

Discussion of scientific literature on binary stars

MESA

INVITED LECTURES

- **Álvaro Sánchez-Monge**

Formation of high-mass stars





Credit: NASA/ESA

Institute of Space Sciences

EXCELENCIA MARÍA DE MAEZTU

EVALUATION

6 ECTS: 54 hours (lectures, hands-on activities and discussions)

Work required to the students

- Class attendance and active participation
- Small exercises to be presented during the course
- Hands-on activities: analysis of ALMA data.
- Running and interpretation of stellar evolution models
- Elaboration of observational proposal
- Discussion of scientific papers



https://campusvirtual.ub.edu

50% of the final mark

- Written examination: short-answer questions

50% of the final mark

Master Thesis Proposals

(https://icc.ub.edu/master_afpc/thesis/astro)

Discovering exotic Galactic transients in time-domain data

Advisor: Nadia Blagorodnova



Modeling the molecular emission in protostellar jets

Advisor: Josep Maria Masqué

OTHER IDEAS

Characterizing Proplyds in the Orion Nebula Cluster with VOLS survey data

Advisor: Gemma Busquet

Exploring the variability of Yellow Supergiants with large time-domain surveys

Advisor: Nadia Blagorodnova

CN(2-1) observations in Orion: searching the total magnetic field in high-mass filaments

Advisor: Gemma Busquet